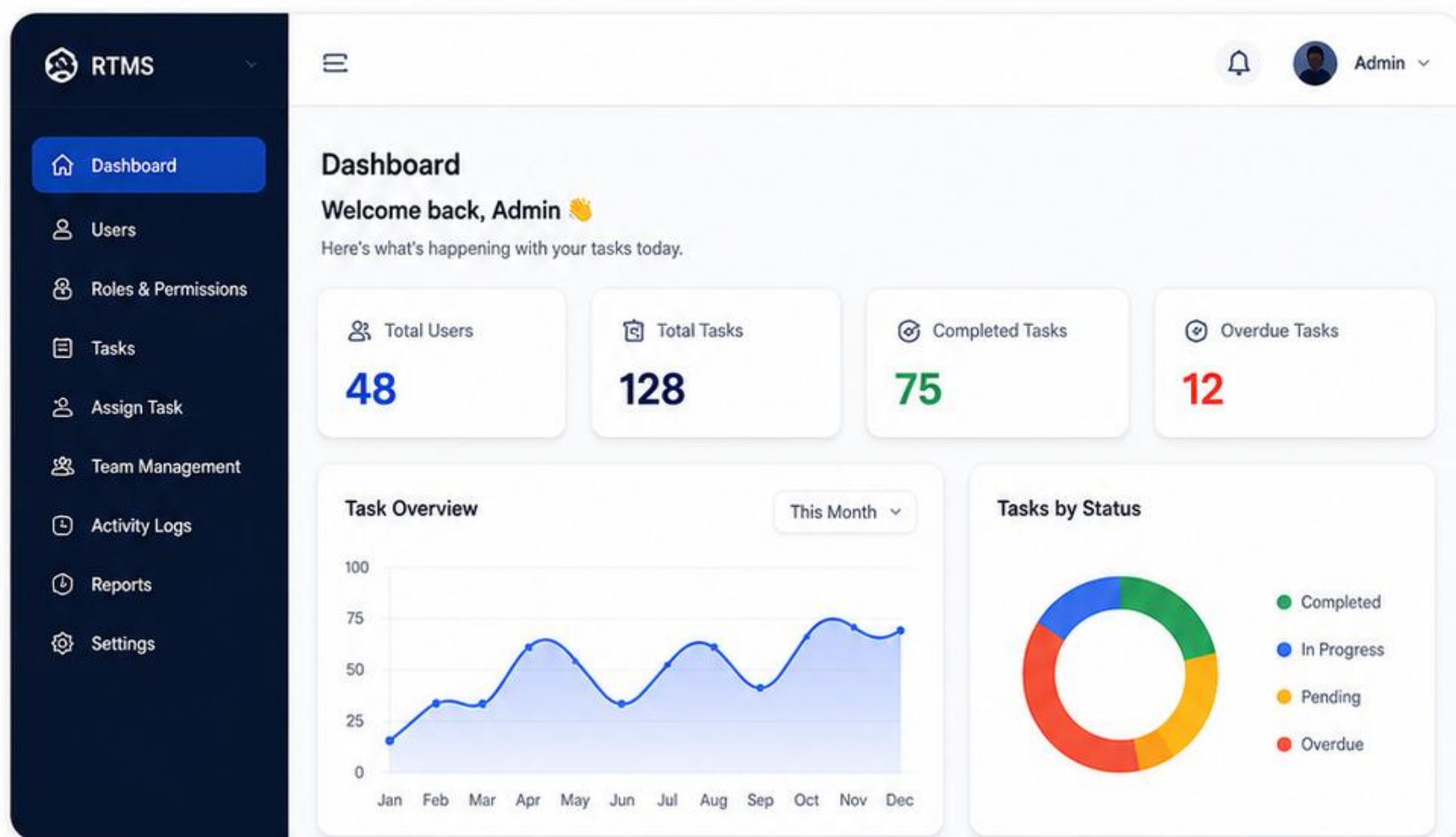


ROLE-BASED TASK MANAGEMENT SYSTEM

• A Full Stack MERN Application •



TECHNOLOGY STACK



Developed By

Charvin

Full Stack Developer



React.js



Node.js



Express.js



MongoDB



Redux Toolkit



JWT Authentication



Node Cron



Nodemailer



Excel Report Generation

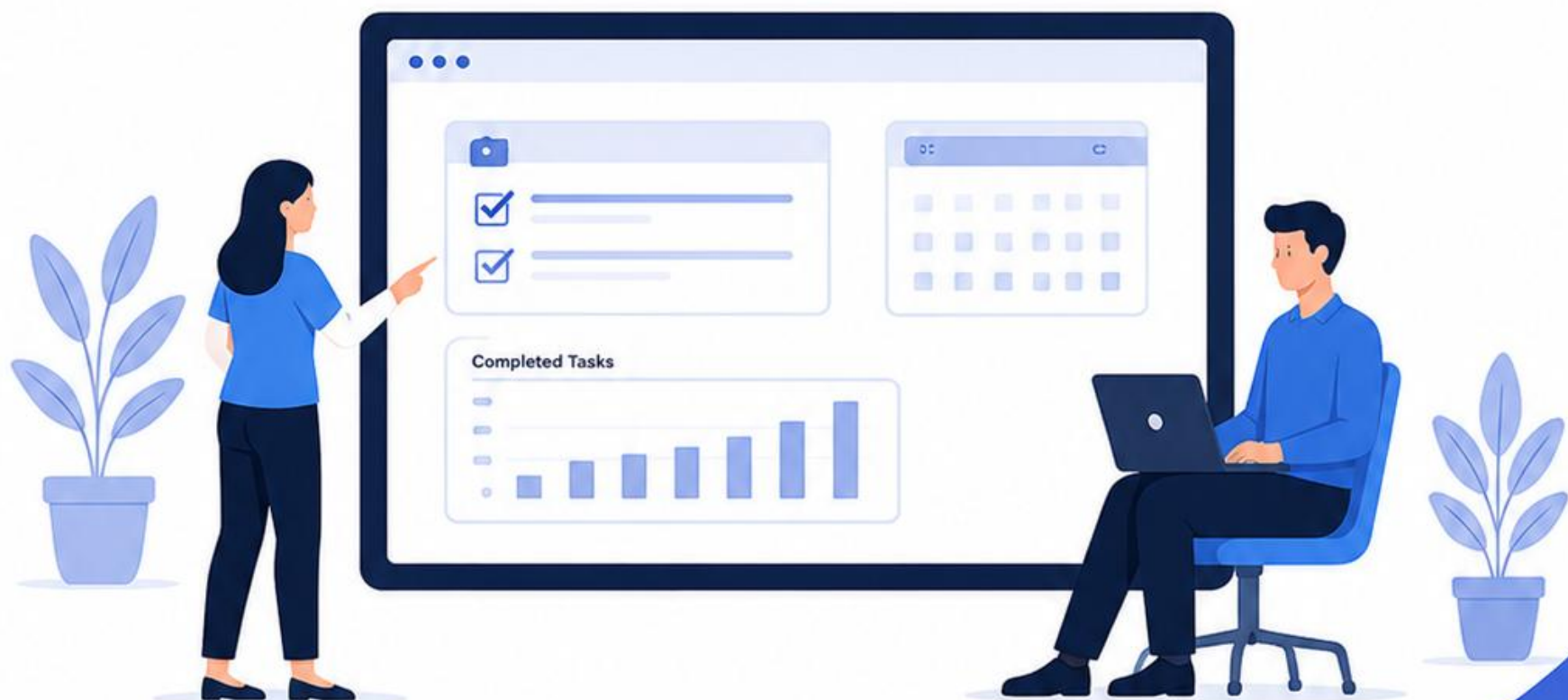


PROJECT OVERVIEW

Role-Based Task Management System is a full-stack MERN application designed to manage employees, team leaders, managers, and administrators through a secure role-based workflow.

The system provides an efficient way to **create, assign, track** and **manage** tasks across different levels of an organization. It ensures accountability, transparency and timely task completion through automation, email notifications, activity logs and periodic reporting.

With features like role-based dashboards, task filtering, overdue task alerts, and automatic monthly Excel report generation, the system streamlines productivity and enhances management efficiency.





KEY FEATURES



Secure JWT Authentication

Secure login and authentication using JSON Web Tokens.



Email Notifications

Automatic emails for login, logout, task assignment, overdue tasks, reset password and more.



Role-Based Access Control

Four user roles: Admin, Manager, Team Leader, Employee with different permissions.



Activity Logs

Detailed logs for logins, logouts and important user actions.



Task Management

Create, update, delete and track tasks with status and priority.



Overdue Task Alerts

Node Cron checks everdue tasks and sends email alerts to respective employees.



Task Assignment

Assign tasks to employees and team leaders efficiently.



Monthly & Yearly Filters

Filter tasks and reports by month and year for better insights.



Password Management

Forgot Password, Reset Password and Change Password functionality.



Automatic Excel Reports

Node Cron automatically generates one month Excel report.



TECHNOLOGY STACK & ARCHITECTURE

TECHNOLOGY STACK



React.js

Frontend Library



Redux Toolkit

State Management



Node.js

Runtime Environment



Express.js

Web Framework



MongoDB

Database



JWT

Authentication



Nodemailer

Email Service



Node Cron

Task Scheduling



ExcelJS

Excel Generation



Bcrypt.js

Password Hashing

ARCHITECTURE



Frontend

React.js



Backend API

Node.js
Express.js



Database

MongoDB



Automation

Node Cron



Services

Email & Excel
Report



The system follows a modular and scalable architecture. The frontend communicates with the backend API, which interacts with the MongoDB database. Node Cron handles scheduled tasks like overdue alerts and monthly report generation, while Nodemailer and ExcelJS manage email notifications and report creation.



SYSTEM DEMONSTRATION

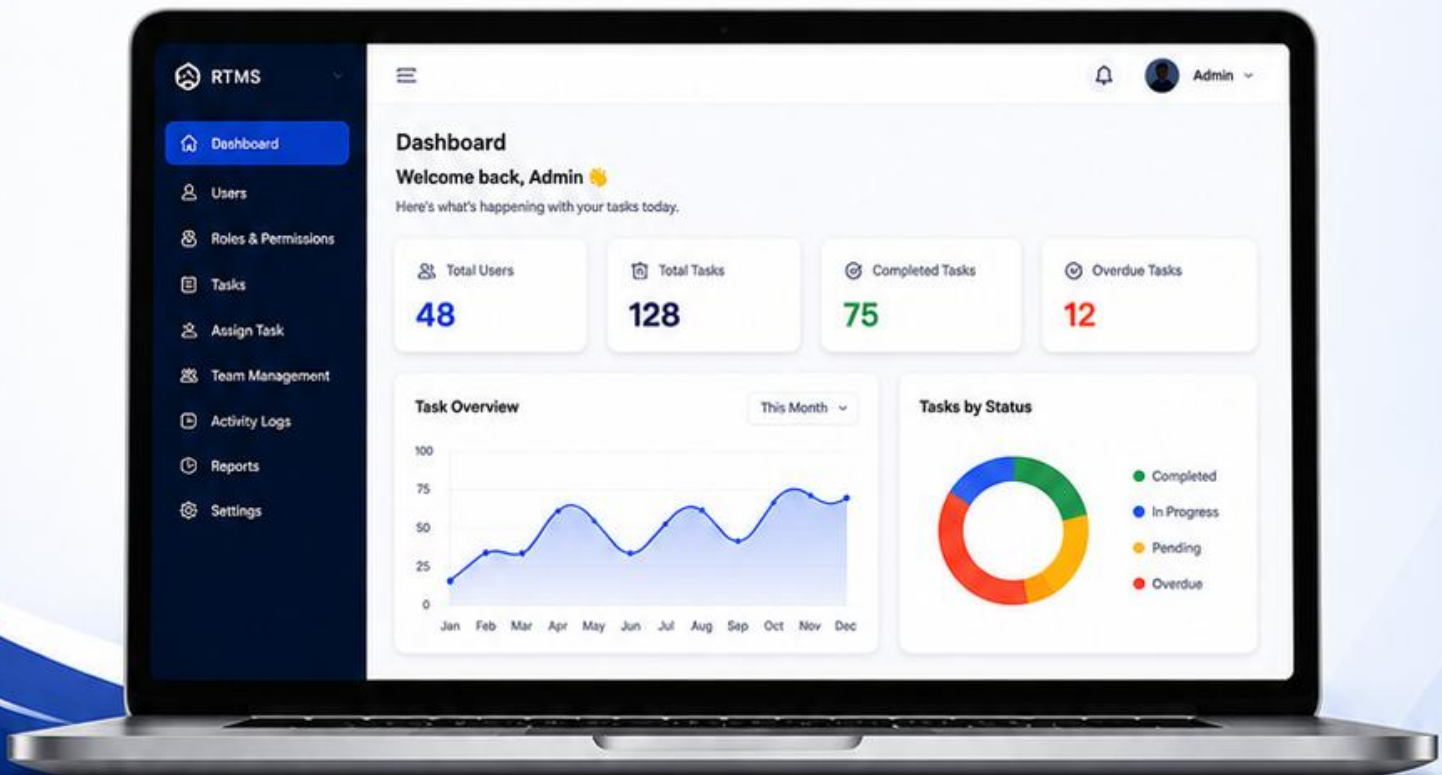
Application Screenshots & Workflow



The following pages demonstrate the user interface, functional modules, and workflow of the **Role-Based Task Management System**.



Screenshots include authentication, role management, task assignment, email notifications, activity logs, automation features, and reporting modules.



“ Explore the complete workflow and features through the following screenshots. ”

✦ Smart Task Management Platform

Manage Your Tasks Like A Professional

Organize projects, boost productivity and collaborate with your team using our modern and powerful task management system.

Get Started

Login

Powerful Features

Everything you need to manage work professionally.



Fast Workflow

Manage your daily tasks with smooth and optimized performance.



Track Progress

Monitor project progress and productivity in real-time.



Team Collaboration

Collaborate with your teammates efficiently and professionally.

Why Choose TaskFlow?

TaskFlow helps developers, startups and teams manage projects with a modern workflow, beautiful interface and productivity-focused tools. Designed for speed, simplicity and scalability.



Ready To Boost Productivity?

Create your account and start managing smarter today.

Create Account



AUTHENTICATION MODULE

User Registration & Secure Login



USER SIGNUP

Signup

Enter name

Enter email

Enter password

Signup

if you have an account so please login
[login](#)



USER LOGIN

Login

Enter email

Enter password

Login

[forget password](#)



Secure Authentication System

The application implements a secure authentication system using JWT (JSON Web Tokens) and encrypted password storage. Users can create accounts, securely log in, and access role-specific dashboards based on their assigned permissions. Authentication middleware protects sensitive routes and ensures only authorized users can access system resources.

KEY SECURITY FEATURES



Secure User
Registration



JWT-Based
Authentication



Encrypted
Password Storage



Protected
Routes



Session
Management

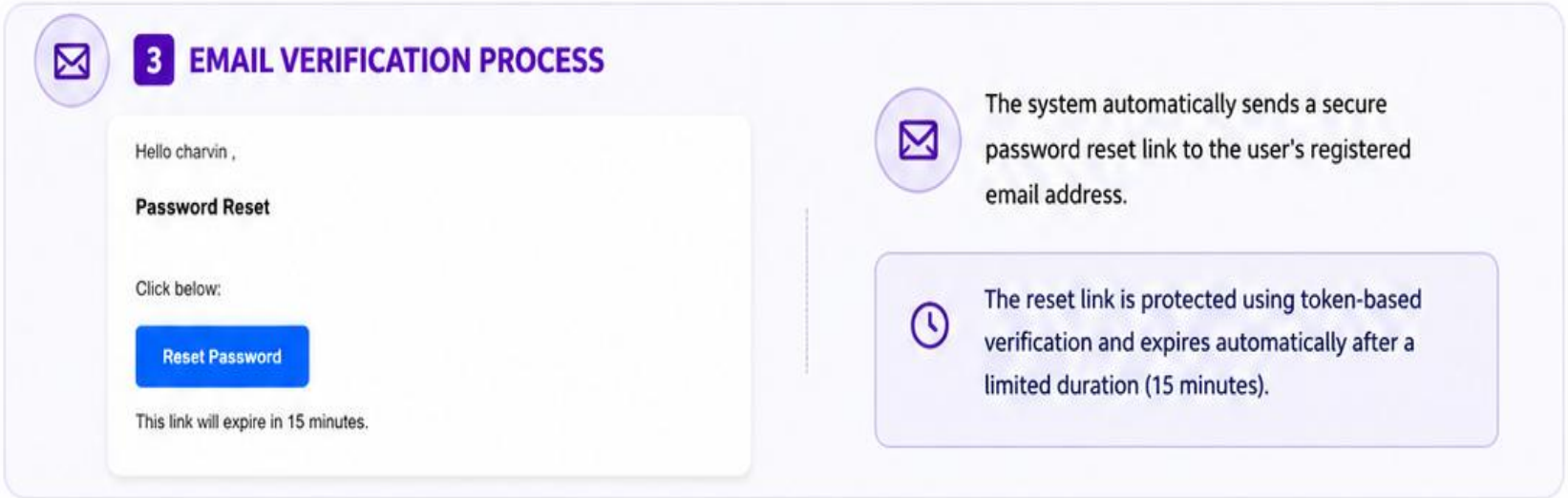
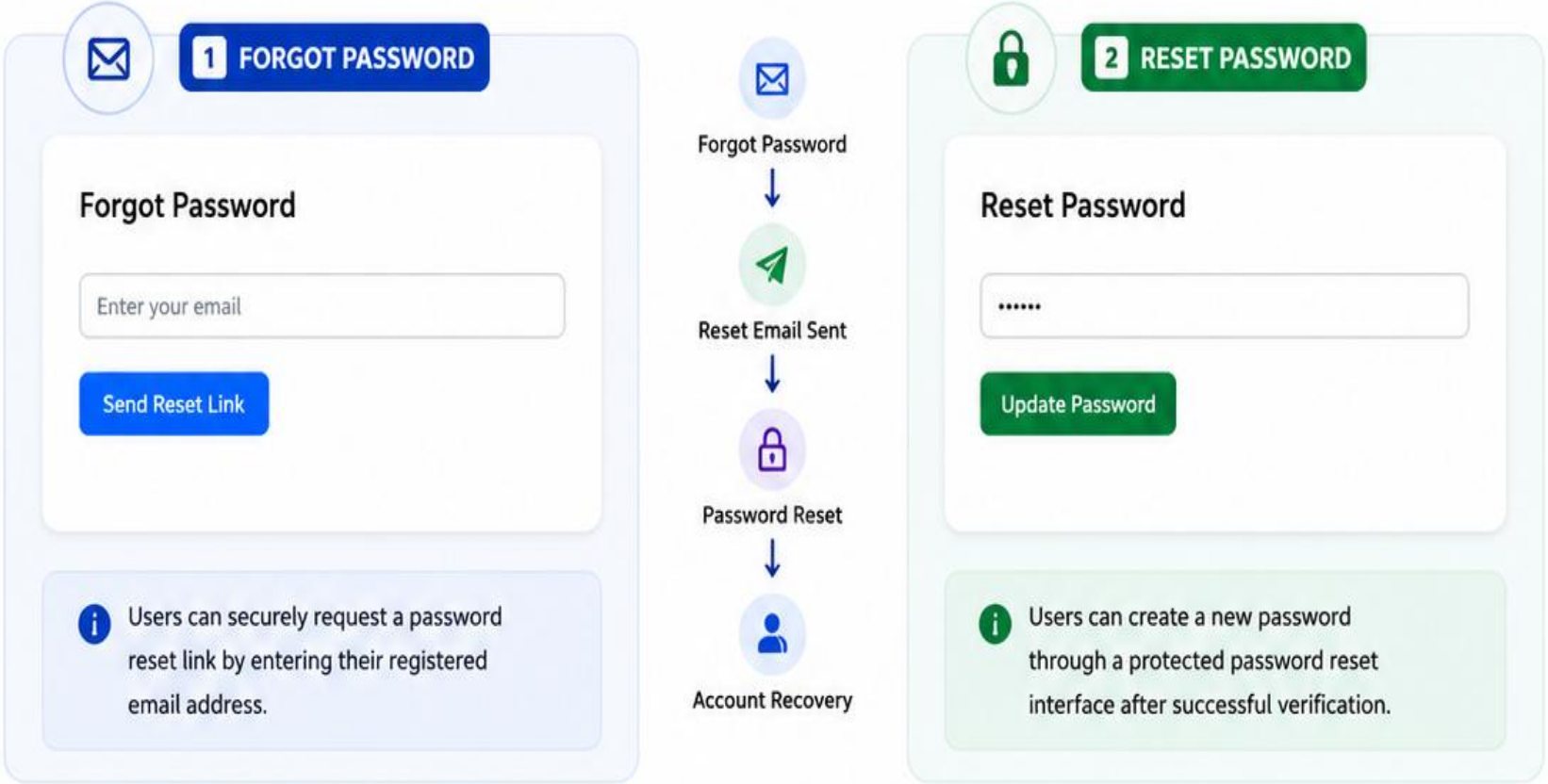


Role-Based
Access Control



PASSWORD RECOVERY MODULE

Secure Password Reset Workflow



SECURITY FEATURES



Email-Based Verification

Ensures only valid users can request password reset.



Time-Limited Reset Links

Reset links expire automatically after 15 minutes.



Secure Token Validation

Token-based verification prevents unauthorized access.



Protected Password Update

Ensures secure password update through validation.



Enhanced Account Security

Complete security measures to protect user accounts.

“ A secure password recovery workflow helps protect user accounts while ensuring seamless account access restoration. ”



ADMINISTRATOR CONTROL PANEL

Overview

The Administrator is the highest authority in the Task Management System. This role is responsible for managing employees, monitoring tasks, maintaining system data, and ensuring smooth workflow across the organization.

1 ADMIN DASHBOARD (OVERVIEW)

Task Manager

john doe

admin

Settings

Logout

Task Manager

Profile

Dashboard

Employees

Tasks

Logs

Admin Dashboard

Total Task
7

Completed
1

In progress
0

Pending
0

Overdue
6

Filter Tasks

April

2026

Apply Filter

Task Distribution

Completed

In Progress

Pending

Overdue

Recent Tasks

Title	Status	Priority
create a web app	overdue	high
create a temp web app	overdue	high
create a temp 2 web app	overdue	high
create a simple crud	overdue	high
create good food delivery web app	overdue	high
create a good library management web pp	overdue	high
A checking task..	completed	high

2 EMPLOYEES MANAGEMENT

Task Manager

john doe

admin

Settings

Logout

Task Manager

Profile

Dashboard

Employees

Tasks

Logs

All Employees

Name	Email	Role	Action
john doe	john@gmail.com	Admin	Delete
adam	adam@gmail.com	Manager	Delete
bob	bob@gmail.com	TL	Delete
harry	harry@gmail.com	Employee	Delete
		Employee	Delete
		Employee	Delete
roy	roy@gmail.com	Employee	Delete
mickay	mick@gmail.com	Employee	Delete
jack	jack@gmail.com	Employee	Delete
jey	jey@gmail.com	Employee	Delete

3 TASK MANAGEMENT

Task Manager

john doe

admin

Settings

Logout

Task Manager

Profile

Dashboard

Employees

Tasks

Logs

Filter Tasks

April

2026

Apply Filter

All Tasks

Title	Status	Priority	Start	End	Action	Edit
create a web app	overdue	high	4/15/2026, 9:10:00 PM	4/15/2026, 8:15:00 PM	Delete	Edit
create a temp web app	overdue	high	4/15/2026, 3:06:00 PM	4/15/2026, 3:10:00 PM	Delete	Edit
create a temp 2 web app	overdue	high	4/15/2026, 3:15:00 PM	4/15/2026, 3:20:00 PM	Delete	Edit
create a simple crud	overdue	high	4/20/2026, 8:10:00 PM	4/20/2026, 8:20:00 PM	Delete	Edit
create good food delivery web app	overdue	high	4/20/2026, 8:20:00 PM	4/20/2026, 8:25:00 PM	Delete	Edit
create a good library management web app	overdue	high	4/20/2026, 8:30:00 PM	4/20/2026, 8:40:00 PM	Delete	Edit
A checking task..	completed	high	4/21/2026, 1:08:00 PM	4/21/2026, 1:30:00 PM	Delete	Edit

4 PROFILE MANAGEMENT (VIEW & UPDATE)

Task Manager

john doe

admin

Settings

Logout

Task Manager

Profile

Dashboard

Employees

Tasks

Logs

My Profile

john doe

john@gmail.com

User ID:

Joined: 4/12/2026, 5:34:49 PM

Edit

Edit User

john doe

john@gmail.com

Cancel

Update

Admin can view and update personal profile information such as name and email to keep account details accurate and up to date.



The Administrator Dashboard provides complete control over employees, tasks, user profiles, and system monitoring features to ensure smooth and efficient operations.

MANAGER MODULE



Manage Team Leaders, Employees & Monitor Performance



OVERVIEW

The Manager module enables managers to monitor team performance, assign Team Leaders to employees, manage profiles, and access activity logs for better decision-making and team management.

KEY FEATURES



View Dashboard Statistics

Get quick insights on Employees, TLs, Managers and performance.



Monitor Team Leaders & Performance

Track TL performance and their assigned tasks.



Team Size Distribution

See the number of employees under each TL.



Assign TL to Employee

Assign employees to Team Leaders easily.



Profile Management

View and update personal profile information.



Activity Logs

Access system activity logs and monitor important actions.

1 MANAGER DASHBOARD

Task Manager

adam manager

Settings

Logout

Task Manager

Profile

Dashboard

Assign TL

Logs

Manager Dashboard

Total Employees
7Total TL
1Total Managers
1

Team Leaders

Name	Email
bob	bob@gmail.com

TL Performance

Name	Total	Completed	Pending	In Progress
bob	12	1	1	2

Team Size per TL

Name	Email	Team Size
bob	bob@gmail.com	5



Dashboard gives a complete overview of team structure, TL performance and team size.

2 ASSIGN TL TO EMPLOYEE (INTERFACE)

Task Manager

adam manager

Settings

Logout

Profile

Dashboard

Assign TL

Logs

Assign TL to Employee

Select TL

Select TL

Select Employee

Select Employee

Assign TL



Select a Team Leader and Employee and click on Assign TL to create the mapping.

3 TL & EMPLOYEE SELECTION (EXAMPLE)

Task Manager

adam manager

Settings

Logout

Profile

Dashboard

Assign TL

Logs

Assign TL to Employee

Select TL

bob

Select Employee

harry (employee)

Assign TL

Employee 'harry (employee)' successfully assigned to TL 'bob'.



After selecting TL and Employee, the employee is successfully assigned to the selected TL.

4 PROFILE MANAGEMENT (VIEW & UPDATE)

Task Manager

adam manager

Settings

Logout

Profile

Dashboard

Assign TL

Logs

My Profile



Email : adam@gmail.com

User ID : 682f8e8b1e0a2f1c9d123abc

Joined : 4/13/2026, 10:48:02 AM

Edit

Edit User

Name

adam

Email

adam@gmail.com

Cancel

Update



View profile information and update name and email. Click Update to save changes.

5 ACTIVITY LOGS

Task Manager

adam manager

Settings

Logout

Profile

Dashboard

Assign TL

Logs

Activity Logs

User	Action	Description	Date & Time
adam	LOGIN	Manager logged in to the system	4/13/2026, 10:45:02 AM
adam	ASSIGN_TL	Assigned TL 'bob' to employee 'harry'	4/13/2026, 10:47:35 AM
adam	UPDATE_PROFILE	Updated profile information	4/13/2026, 11:00:12 AM
adam	LOGOUT	Manager logged out	4/13/2026, 11:25:08 AM



View system activity logs to monitor important actions performed in the system.



MANAGER MODULE BENEFITS



Centralized Team Monitoring



Better Team Management



Performance Tracking



Easy TL & Employee Management



Transparency & Accountability

“ Empowering Managers to Lead, Monitor and Achieve More. ”

TEAM LEADER (TL) MODULE

Create Tasks. Assign Work. Track Progress. Lead Your Team.



1 OVERVIEW

The Team Leader (TL) module helps TLs manage their team effectively by creating tasks, assigning them to team members, tracking progress, and monitoring task status. TLs can also manage their profile and access activity logs.

2 KEY FEATURES



Create Tasks

Create new tasks with priority and timeline.



Assign Tasks

Assign created tasks to team members.



Manage Tasks

View all created tasks and perform edit or delete.



Track Team Progress

Filter tasks by month and year and see status distribution.



Profile Management

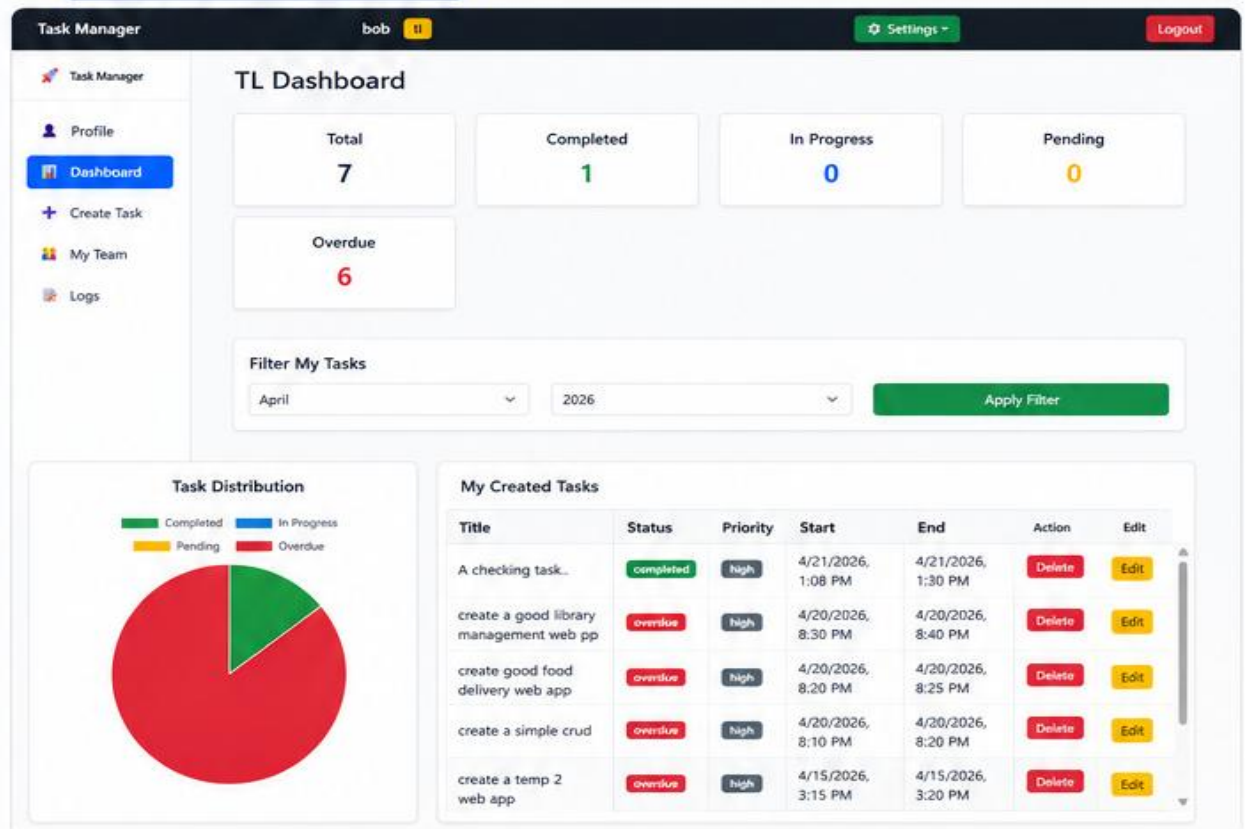
View and update personal profile information.



Activity Logs

View system activity logs and important actions.

3 TL DASHBOARD (OVERVIEW)



Dashboard provides TLs a quick view of task statistics, distribution, and list of tasks they created.

4 CREATE TASK & ASSIGN TO TEAM MEMBER

Task Manager bob

Create Task:

- Title: Enter task title
- Description: Enter task description
- Select Priority: [Dropdown]
- Start Date: dd-mm-yyyy --:--
- End Date: --:--

Assign Task:

- Select Task: [Dropdown]
- Select Employee: [Dropdown]
- Assign Task: [Button]



TL can create tasks and assign them to team members with start/end date and priority.

5 MY TEAM (TEAM MEMBERS)

Task Manager bob

My Team

Name	Email
charvin	sh...
roy	roy@gmail.com
mickey	mic@gmail.com
jack	jack@gmail.com
jeey	jeey@gmail.com



TL can view all team members assigned under them.

6 PROFILE & UPDATE

Task Manager

My Profile:

- Profile: bob
- Email: bob@gmail.com
- User ID: [ID]
- Joined: 4/13/2026, 10:49:03 AM
- Edit: [Button]

Edit User:

- Name: bob
- Email: bob@gmail.com
- Cancel: [Button]
- Update: [Button]



TL can view and update their profile information such as name and email.

7 ACTIVITY LOGS

Task Manager

Activity Logs

User	Action	Description	Date & Time
bob	LOGIN	TL logged in to the system	4/18/2026, 10:20:15 AM
bob	CREATE_TASK	Created task "A checking task..."	4/18/2026, 10:21:05 AM
bob	ASSIGN_TASK	Assigned task "A checking task..." to mickey	4/18/2026, 10:22:40 AM
mickey	UPDATE_TASK	Updated status to Completed	4/18/2026, 10:30:12 AM
bob	LOGOUT	TL logged out	4/18/2026, 10:45:30 AM



TL can access activity logs to monitor important actions performed in the system.

BENEFITS OF TL MODULE



Efficient Task Management



Better Team Coordination



Real-time Progress Tracking



Improved Productivity & Accountability



Complete Visibility of Team Work



Empowering Team Leaders to Manage, Assign and Track Work Effectively.



EMPLOYEE MODULE



MODULE 06
EMPLOYEE MODULE

View Tasks, Track Progress & Update Status



OVERVIEW

The Employee module allows employees to view their assigned tasks, track their progress, update task status, and manage their personal profile. It ensures a simple, focused, and secure experience with access to only their own data.



KEY FEATURES



Employee Dashboard

Get an overview of your tasks and performance using statistics and charts.



My Tasks & Status Update

View all assigned tasks and update their status (Pending, In Progress, Completed).



Profile Management

View and update your personal information such as name and email.



Secure Access

Employees can access only their own tasks and profile information for better security.



EMPLOYEE DASHBOARD (OVERVIEW)

Task Manager

charvin

employee

Settings

Logout

Profile

Dashboard

My Tasks

Employee Dashboard

Total
11

Completed
1

In Progress
0

Pending
0

Filter My Tasks

April

2026

Apply Filter

My Task Distribution

Completed In Progress
Pending Overdue



Dashboard provides a quick overview of your tasks and performance. Use month and year filter to analyze your task progress.



MY TASKS & STATUS UPDATE

Task Manager

charvin

employee

Settings

Logout

Profile

Dashboard

My Tasks

My Tasks

Title	Status	Priority	Start Time	End Time
create a web app	Pending	high	4/15/2026, 9:10:00 PM	4/15/2026, 8:15:00 PM
create a temp 2 web app	Pending	high	4/15/2026, 3:15:00 PM	4/15/2026, 3:20:00 PM
create good food delivery web app	Pending	high	4/20/2026, 8:20:00 PM	4/20/2026, 8:25:00 PM
create a good library management web pp	Pending	high	4/20/2026, 8:30:00 PM	4/20/2026, 8:40:00 PM
A checking task..	Completed	high	4/21/2026, 1:08:00 PM	4/21/2026, 1:30:00 PM



View all tasks assigned to you and update the status as Pending, In Progress or Completed.

Pending

In Progress

Completed



PROFILE MANAGEMENT

Task Manager

charvin

employee

Settings

Logout

Profile

Dashboard

My Tasks

My Profile



charvin

employee

Email: sha

User ID: 6

Joined: 4/15/2026, 12:26:12 PM

Edit

Edit User

Name

charvin

Email

sha

Cancel

Update



Employees can view and update their personal profile information.



Update Name



Update Email Address



Profile management helps keep your account information accurate and secure.



BENEFITS OF EMPLOYEE MODULE



Personalized Task View



Easy Progress Tracking



Quick Status Updates



Simple & Clean Interface



Improved Productivity



Focused & Secure Experience



Empowering Employees to Stay Focused, Track Progress and Deliver Results.





Track All System Activities & Maintain Full Transparency

1 OVERVIEW

The Activity Log module records every important action performed in the system by Admins, Managers, TLs and Employees. It helps in monitoring user activities, maintaining accountability, ensuring security and providing a complete audit trail of all operations.

2 LOGGED ACTIVITIES INCLUDE

- Login / Logout Activities**
Track user login and logout activities.
- Task Management**
Task created, updated, assigned and status changes.
- User Management**
User / Role created, updated and deleted.
- Profile Updates**
Profile information updated by users.
- TL Assignments**
Team Leader assigned to employees.
- System Events**
Other important system level actions.

3 ACTIVITY LOGS (SYSTEM ACTIVITY OVERVIEW)

Activity Logs					
Export Excel Filters					
No	Action	Task	Employee	Role	Date & Time
1	Logged in	N/A	adam	manager	6/18/2026, 3:50:08 PM
2	Logged in	N/A	john doe	admin	6/18/2026, 3:49:43 PM
3	Logged out	N/A	mickey	employee	6/18/2026, 3:49:31 PM
4	Logged in	N/A	mickey	employee	6/18/2026, 3:49:20 PM
5	Logged in	N/A	john doe	admin	6/18/2026, 3:48:35 PM
⋮					
30	Assigned task to charvin	shopify web app	bob	tl	4/15/2026, 12:31:14 PM
31	Created a task	shopify web app	bob	tl	4/15/2026, 12:29:54 PM
32	Updated status to in_progress	Build API with user crud	harry	employee	4/14/2026, 2:17:34 PM
33	Assigned task	Build API with user crud	bob	tl	4/14/2026, 2:16:09 PM
34	Created a task	Build API with user crud	bob	tl	4/14/2026, 2:02:09 PM
Showing 1 to 10 of 34 entries					
« 1 2 3 4 »					

Activity logs capture all important actions performed in the system. Use filters to search by action type, user, role or date range. You can also export all logs in Excel format.

4 LOG CATEGORIES (EXAMPLES)

Login Activities

- adam logged in
- john doe logged in
- mickey logged in
- harry logged in

Logout Activities

- mickey logged out
- john doe logged out
- harry logged out
- bob logged out

Task Activities

- Task created
- Task assigned
- Status updated
- Task deleted

User Activities

- User created
- User updated
- Role changed
- User deleted

TL Assignments

- TL assigned
- TL removed
- TL updated
- TL re-assigned

System Events

- Profile updated
- Password changed
- Settings updated
- Other actions

5 BENEFITS OF ACTIVITY LOGS)

Security Monitoring

Helps detect suspicious activities.

Audit Trail

Maintain complete history of user actions.

User Accountability

Ensure responsibility and transparency.

System Transparency

Increase trust with clear operation logs.

Easy Troubleshooting

Quickly identify and resolve issues.

Compliance Ready

Useful for reporting and compliance.

Automated Email Alerts for User Authentication,
Task Management & Security

OVERVIEW

The Email Notification System automatically sends important notifications to users based on different events occurring within the Task Management System. It improves communication, enhances security, and keeps users informed about account activities and task updates.

KEY FEATURES

- Login Notification**
Email sent when user successfully login.
- Logout Notification**
Email sent when user successfully logout.
- Task Assignment Email**
Email sent when a new task is assigned.
- Overdue Task Reminder**
Email sent when task becomes overdue.
- Password Reset Link**
Email with secure link to reset password.
- Password Changed Confirmation**
Email sent after password changed successfully.

1 LOGIN ALERT

From: Task Manager System <noreply@taskmanager.com>
To: charvin (Private email address)
Subject: Login Alert

Login Alert

Hello charvin ,
You are login on our website ,
login detected at
16/4/2026, 11:23:29 am

This is an automated email. Please do not reply to this email.

2 LOGOUT ALERT

From: Task Manager System <noreply@taskmanager.com>
To: charvin (Private email address)
Subject: Logout Alert

Logout Alert

Hello charvin ,
You have successfully logged out.
logout detected at
16/4/2026, 11:23:48 am

This is an automated email. Please do not reply to this email.

3 NEW TASK ASSIGNED

From: Task Manager System <noreply@taskmanager.com>
To: charvin (Private email address)
Subject: New Task Assigned

New Task Assigned

Hello charvin ,
You have been assigned a new task:
temp task for checking api
Please login to the system for more details.

This is an automated email. Please do not reply to this email.

4 TASK OVERDUE REMINDER

From: Task Manager System <noreply@taskmanager.com>
To: charvin (Private email address)
Subject: Task Overdue

Task Overdue

Hello charvin ,
Your task **"temp task for checking api"** is overdue. Please complete it as soon as possible.

This is an automated email. Please do not reply to this email.

5 RESET PASSWORD LINK

From: Task Manager System <noreply@taskmanager.com>
To: charvin (Private email address)
Subject: Password Reset

Password Reset

Hello charvin ,
Click the button below to reset your password. This link will expire in 15 minutes.

Reset Password

This is an automated email. Please do not reply to this email.

6 PASSWORD CHANGED

From: Task Manager System <noreply@taskmanager.com>
To: charvin (Private email address)
Subject: Password Changed Successfully

Password Changed Successfully

Hello charvin ,
Your account password has been changed successfully.
If you made this change, no further action is required.
If you did NOT change your password, please click on forget password & reset your password immediately or contact support.

This is an automated email. Please do not reply to this email.

EMAIL NOTIFICATION BENEFITS

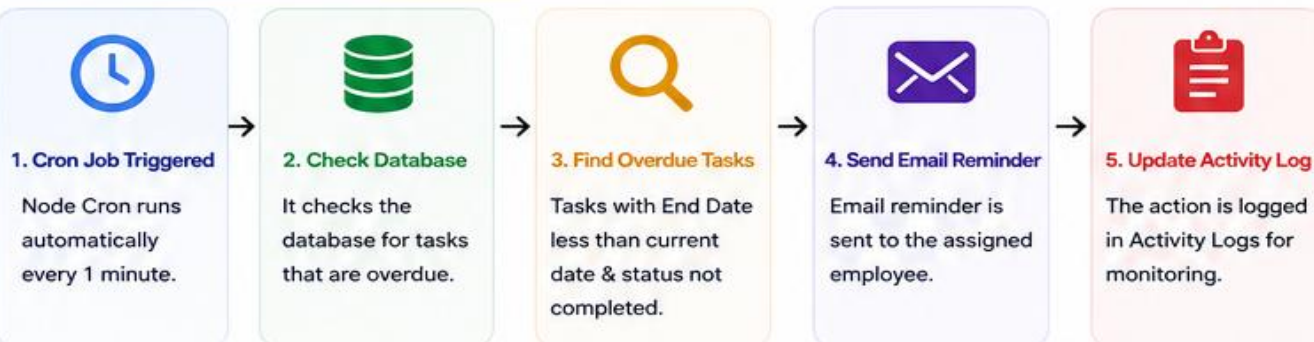
- Instant User Notifications**
Quick updates keep users informed in real-time.
- Improved Account Security**
Login, logout & password alerts help protect user accounts.
- Automated Task Communication**
Task assignments & reminders ensure better coordination.
- Password Recovery Support**
Secure reset links help users recover access to their accounts.
- Real-Time Email Alerts**
Important events are delivered automatically.
- Better User Experience**
Keeps users engaged and informed about their activities.

Automated Overdue Task Detection & Email Reminder System

OVERVIEW

Node Cron is used to run scheduled tasks automatically in the background. In this system, it checks for overdue tasks at regular intervals and sends reminder emails to the assigned employees automatically.

1 WORKFLOW: OVERDUE DETECTION TO EMAIL REMINDER



This process runs in the background automatically and ensures that overdue tasks are detected and notified without any manual work.

2 CRON JOB IMPLEMENTATION (CODE)

```
1 const cron = require("node-cron");
2 const Task = require("../models/taskModel");
3 const TaskAssignment = require("../models/taskAssignmentModel");
4 const Employee = require("../models/employeeModel");
5 const sendEmail = require("../utils/sendEmail");
6
7 cron.schedule("* * * * *", async () => {
8   console.log("Running cron job...");
9
10  try {
11    const now = new Date();
12    const tasks = await Task.find({
13      end_time: { $lt: now },
14      status: { $ne: "completed" },
15      is_overdue_notified: false,
16    });
17    for (let task of tasks) {
18      // mark task as overdue
19      task.status = "overdue";
20      const assignments = await TaskAssignment.find({
21        task_id: task._id,
22      });
23      for (let assign of assignments) {
24        const employee = await Employee.findById(assign.employee_id);
25        await sendEmail(
26          employee.email,
27          "Task Overdue ⚠️",
28          `Hello ${employee.name}, your task "${task.title}" is overdue. Please complete it as soon as possible.`
29        );
30        // mark as notified
31        task.is_overdue_notified = true;
32        await task.save();
33      }
34    }
35    catch (error) {
36      console.log("Cron error:", error.message);
37    }
38  }
39 });
```

Cron Expression Explained: * * * * * = Every 1 Minute
(Runs at every minute of every hour, every day)

3 SAMPLE OVERDUE EMAIL REMINDER

From Task Manager System <noreply@taskmanager.com>
To charvin (Private email address)
Subject Task Overdue ⚠️



Hello charvin,
Your task "temp task for checking api" is overdue.
Please complete it as soon as possible.

Task Details

Task Title : temp task for checking api
Due Date : 16/04/2026
Priority : High
Status : **Overdue**

This is an automated email. Please do not reply to this email.

TECHNOLOGIES USED

node.js Node.js

node-cron node-cron

Nodemailer

MongoDB

WHY NODE CRON?

- ✓ Automates repetitive tasks
- ✓ No manual intervention
- ✓ Accurate & time-saving
- ✓ Improves productivity
- ✓ Real-time email reminders

4 KEY FEATURES

- Automatic Overdue Detection**
Detects overdue tasks automatically without manual checking.
- Email Reminder Automation**
Sends email reminders to employees for overdue tasks.
- Activity Log Updates**
Every action is recorded in the system for transparency.
- Scheduled Execution**
Cron job runs in the background at regular intervals.
- No Manual Intervention**
Reduces manual work and ensures timely notifications.

5 BENEFITS

- Timely Reminders**
Employees get notified on time about overdue tasks.
- Better Productivity**
Encourages employees to complete tasks before deadlines.
- Improved Accountability**
Automatic logs and emails ensure responsibility and tracking.
- Saves Time**
Eliminates manual checking and sending notifications.
- Enhanced Project Tracking**
Keeps projects on track by reducing delays caused by pending tasks.

Monthly Automated Task Report Export in Excel Format

OVERVIEW

The system automatically generates Excel reports of all tasks every month. The report includes complete task details such as status, priority, dates, and assigned users. Administrators can download and store these reports for analysis, auditing, and record keeping.

AUTOMATION FLOW



CRON SCHEDULE

Runs Every Month
(1st Day of Every Month)
at 00:00 AM

Cron Expression
0 0 1 * * *

TECHNOLOGIES USED



1 AUTOMATICALLY GENERATED EXCEL REPORT (SAMPLE)

Title	Status	Priority	Start Time	End Time	Created By
create a web app	overdue	high	15/4/2026, 9:10:00 pm	15/4/2026, 8:15:00 pm	bob
create a temp web app	overdue	high	15/4/2026, 3:06:00 pm	15/4/2026, 3:10:00 pm	bob
create a temp 2 web app	overdue	high	15/4/2026, 3:15:00 pm	15/4/2026, 3:20:00 pm	bob
create a simple crud	overdue	high	20/4/2026, 8:10:00 pm	20/4/2026, 8:20:00 pm	bob
create good food delivery web	overdue	high	20/4/2026, 8:20:00 pm	20/4/2026, 8:25:00 pm	bob
create a good library manager	overdue	high	20/4/2026, 8:30:00 pm	20/4/2026, 8:40:00 pm	bob
A checking task..	completed	high	21/4/2026, 1:08:00 pm	21/4/2026, 1:30:00 pm	bob

2 DOWNLOAD EXCEL REPORT

Task Reports

Report Name	Month	Generated On	Records	Action
task_report_4_2026.xlsx	April - 2026	01/04/2026 12:00 AM	7	Download

Reports are automatically generated on the 1st day of every month and available for download.

3 CRON JOB CODE (AUTOMATIC REPORT GENERATION)

```
const cron = require("node-cron");
const Task = require("../models/taskModel");
const ExcelJS = require("exceljs");
const fs = require("fs");
const path = require("path");

cron.schedule("0 0 1 * * *", async () => { // Every month on 1st day at 00:00 AM
  try {
    const tasks = await Task.find().populate("created_by", "name");
    const workbook = new ExcelJS.Workbook();
    const worksheet = workbook.addWorksheet("Task Report");

    worksheet.columns = [
      { header: "Title", key: "title", width: 30 },
      { header: "Status", key: "status", width: 15 },
      { header: "Priority", key: "priority", width: 15 },
      { header: "Start Time", key: "start_time", width: 22 },
      { header: "End Time", key: "end_time", width: 22 },
      { header: "Created By", key: "created_by", width: 15 },
    ];

    tasks.forEach(task => {
      worksheet.addRow({
        title: task.title,
        status: task.status,
        priority: task.priority,
        start_time: task.start_time,
        end_time: task.end_time,
        created_by: task.created_by?.name || "N/A"
      });
    });

    const fileName = `task_report_${new Date().getMonth()+1}_${new Date().getFullYear()}.xlsx`;
    const filePath = path.resolve(__dirname, "../reports", fileName);
    await workbook.xlsx.writeFile(filePath);

    console.log("Excel report generated: ", fileName);
  } catch (error) {
    console.log("Cron Error: ", error.message);
  }
});
```

Cron Expression: 0 0 1 * * * → Runs every month on 1st day at 00:00 AM

4 SAMPLE EMAIL WITH ATTACHED REPORT (OPTIONAL)

From: Task Manager System <noreply@taskmanager.com>
To: admin@taskmanager.com
Subject: Monthly Task Report - April 2026

Monthly Task Report

Hello Admin,
Please find attached the automated task report for April - 2026.

Total Records : 7
Generated On : 01/04/2026 12:00 AM

Regards,
Task Manager System

task_report_4_2026.xlsx
7 KB

BENEFITS OF AUTOMATIC EXCEL REPORT GENERATION



Saves Time
No manual work required. Reports generated automatically.



Accurate Data
Real and complete data directly fetched from database.



Easy Analysis
Structured Excel format makes analysis and filtering easy.



Better Tracking
Monthly records help in tracking performance and progress.



Data Backup
Reports can be stored for future reference and auditing.



Professional Reports
Excel reports are professional and easily shareable with others.